

Damage Classification of Steel Frames Using Long Short-Term Memory and Fully Convolutional Network Models

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Abstract. In the field of structural health monitoring (SHM), the application of deep learning models for analyzing time-series data has garnered significant attention. One-dimensional convolutional neural networks (1DCNN) are commonly used but face limitations in effectively handling long datasets. Therefore, this study proposes a novel approach by combining 1DCNN with the Squeeze-and-Excitation (SE) mechanism (SE-1DCNN) and Long Short-Term Memory (LSTM) networks to accurately classify structural damage. This combination leverages the spatial feature extraction and attention mechanism of SE-1DCNN alongside LSTM's capability to process long-term time-series data. The model is trained and evaluated using an experimental dataset collected from a steel frame structure instrumented with multiple accelerometers under various damage scenarios. The proposed SE-1DCNN-LSTM model achieves an accuracy of 96.7% on the training set and 95.3% on the test set, outperforming the traditional 1DCNN-LSTM model. These results confirm that integrating SE-1DCNN and LSTM enhances damage classification accuracy and demonstrates strong potential for real-world SHM applications.

Keywords: One-Dimensional Convolutional Neural Network with Squeeze-and-Excitation Mechanism (SE-1DCNN), Long Short-Term Memory Network (LSTM), Structural Health Monitoring System (SHM).

1 Introduction

In Vietnam, ensuring the safety of transport infrastructure such as bridges increasingly depends on Structural Health Monitoring (SHM) systems. These systems collect real-time data (e.g., acceleration, strain, displacement) to spot damage at an early stage and enable more practical maintenance planning [1].

Although AI methods have recently been introduced to improve SHM accuracy, we have noticed that traditional machine learning still struggles with large, high-dimensional sensor data [2]. Deep learning models provide a more promising direction, especially for handling complex time-series patterns [3]. LSTM networks, in particular, can learn long-term temporal behavior but often face overfitting and high computation demands [4]. Likewise, 1D convolutional neural networks (1DCNNs) extract local

features well but tend to lose global temporal information and struggle with variable-length data [5,6].

To overcome these challenges, we designed a hybrid framework combining 1DCNN with a Squeeze-and-Excitation mechanism and an LSTM module. The SE block improves the model's ability to emphasize important features, and the LSTM module captures temporal dependencies. We then evaluated the proposed SE-1DCNN-LSTM model using the QUGS acceleration dataset collected from a steel frame structure under various damage scenarios.

2 Proposed SE-1DCNN-LSTM Network Model

In this study, we focus on classifying structural damage using acceleration data gathered from a sensor network. Our SE-1DCNN-LSTM architecture leverages both spatial and temporal features in the signals, enabling more accurate identification of damage patterns. Fig. 1 illustrates the overall network design.

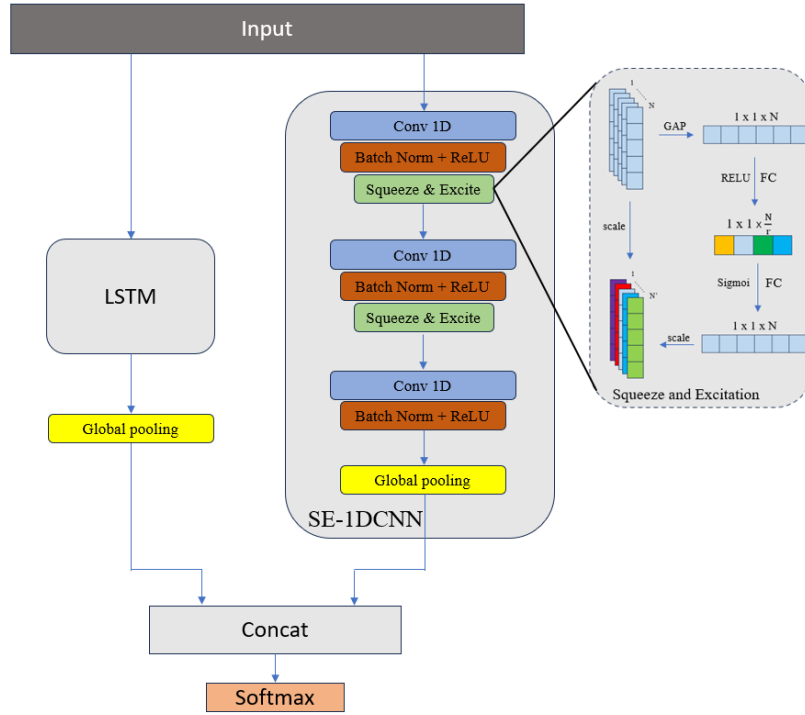


Fig. 1. SE-1DCNN-LSTM Network Model.

The SE-1DCNN-LSTM network processes data through parallel streams. The SE-1DCNN branch captures spatial relationships and extracts local patterns. The LSTM branch learns long-term temporal dependencies. We integrated the Squeeze-and-

Excitation (SE) mechanism [7] within the 1DCNN to help the model prioritize relevant features. This mechanism allows the network to adaptively assign weights to different data channels, focusing attention on signals most critical for damage detection

The SE block improves the model's performance by two key steps: the "squeeze" step compresses spatial information into a global representation for each channel, and the "excitation" step generates attention weights to highlight important channels while suppressing less relevant ones [8]. As a result, SE not only improves feature discrimination but also minimizes noise, allowing the model to focus on significant signals related to damage. The outputs from the SE-1DCNN and LSTM branches are combined and fed into a classification layer to predict the type of damage. The model is optimized through an automated hyperparameter search, fine-tuning factors such as the number of convolutional blocks, number of filters, and dropout rate to achieve the highest performance.

3 MethodologyApplication

3.1 Introduction to the dataset

This study utilizes the QUGS acceleration dataset [9]. The steel frame structure of QUGS (Fig. 2) consists of 30 joints between the main beams and secondary beams, corresponding to 30 locations where 30 acceleration sensors are installed at the joints of the main beams. Damage scenarios are simulated by loosening bolts at these joints.

In this study, we used the first 100,000 data points of the provided dataset. A sample of the utilized dataset is shown in Fig. 3. To enhance the data and improve the model's learning capability, we applied data augmentation techniques to increase the number of samples and segmented the original dataset. A sample of the augmented and reshaped dataset is shown in Fig. 4.

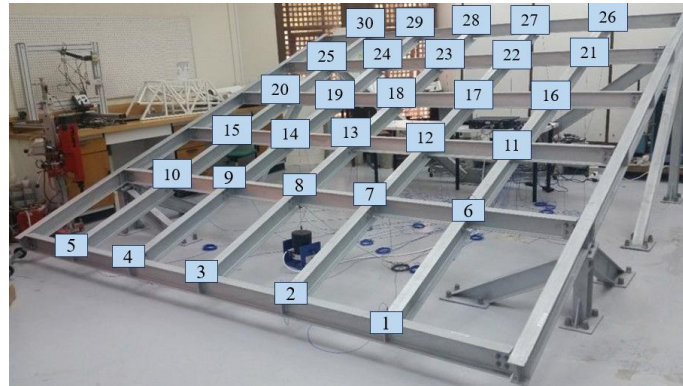


Fig. 2. Steel frame structure and measurement point names.

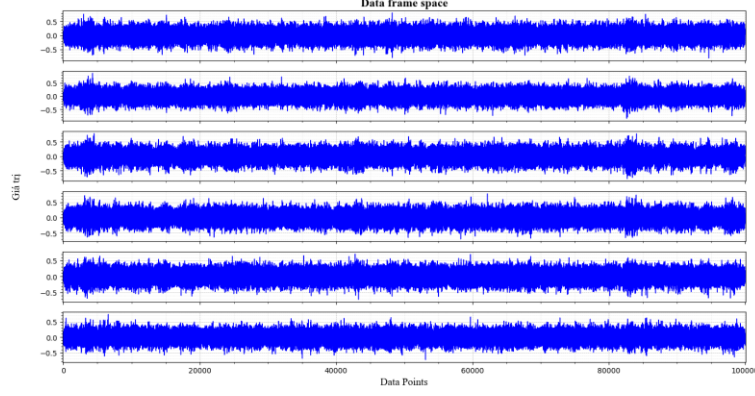


Fig. 3. Utilized Dataset.

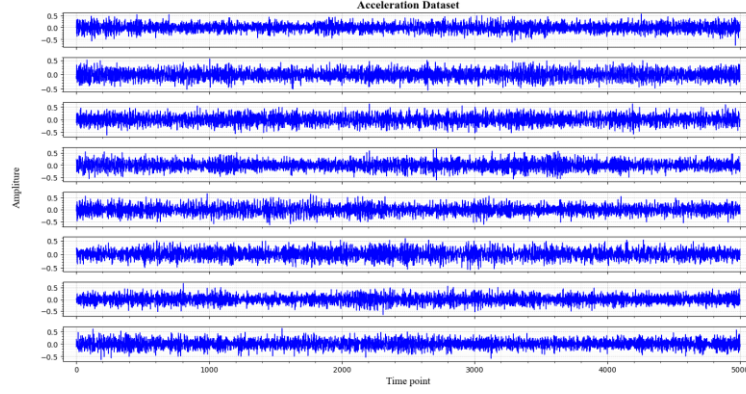


Fig. 4. A sample in the augmented and reshaped dataset.

3.2 Training and Result Evaluation

Fig. 5 and 6 illustrate the accuracy of the two models, 1DCNN-LSTM and SE-1DCNN-LSTM, during the training and testing phases on the steel frame structure dataset. Fig. 5 shows that both models achieve near-perfect accuracy on the training set after just a few initial epochs. Regarding the test set accuracy, the 1DCNN-LSTM model starts with an accuracy of approximately 0.81 and gradually increases to around 0.95. During testing, the 1DCNN-LSTM model experiences a decline in accuracy before stabilizing at 0.935, indicating issues with overfitting and limited generalization on the test data, which the SE-1DCNN-LSTM model does not encounter. In contrast, the SE-1DCNN-LSTM model begins with a slightly lower accuracy of about 0.76 but maintains consistently high and stable accuracy throughout the testing phase. This model generalizes better and is less prone to overfitting, and achieves superior performance in both the training and testing phases.

Fig. 7 and 8 compare the accuracy and loss of the two models, 1DCNN-LSTM and SE-1DCNN-LSTM, on the training and test sets. On the training set, the 1DCNN-

LSTM model achieves an accuracy of 0.935 and a loss of 0.302, while the SE-1DCNN-LSTM model attains a higher accuracy of 0.967 and a lower loss of 0.175, indicating that SE-1DCNN-LSTM is more effective in learning the features of the validation set. On the test set, the 1DCNN-LSTM model achieved an accuracy of 0.926 with a loss of 0.328, while the SE-1DCNN-LSTM model achieved a higher accuracy of 0.953 and a lower loss of 0.242. These results indicate that the SE-1DCNN-LSTM model generalizes better and is less prone to overfitting than the 1DCNN-LSTM model.

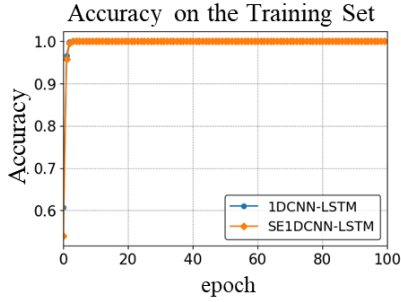


Fig. 5. Accuracy on the Training Set.

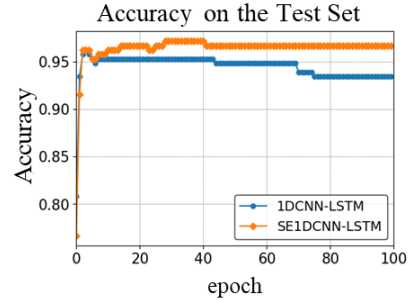


Fig. 6. Accuracy on the Test Set.

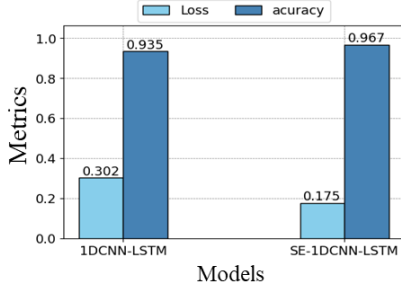


Fig. 7. Loss and Accuracy on the Training Set.

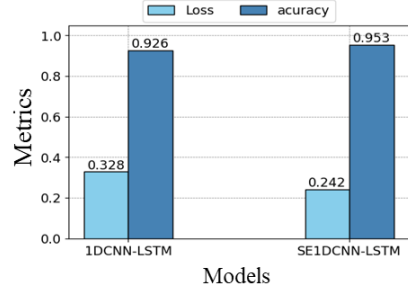


Fig. 8. Loss and Accuracy on the Test Set.

4 Conclusion

In this study, we proposed a deep learning architecture that combines a one-dimensional convolutional neural network (1DCNN), a Squeeze-and-Excitation mechanism, and an LSTM module to classify structural damage from acceleration data. We validated the model using data from a steel frame under various damage scenarios. The results demonstrate that the proposed architecture achieves higher accuracy and improved generalization compared to a conventional 1DCNN-LSTM model.

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